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System and methods for detecting a copy of a deep neural network in a hearing device

Abstract

System and methods are presented for detecting an authentic copy of a deep neural network in a device without reverse engineering the device. For example, the device can be a hearing device. A method is presented that includes providing audio signals to a deep neural network located in a hearing device, receiving signals from the deep neural network that are triggered by the transmitted audio signals, and determining whether the received signals were the expected signals. Specifically, the system can determine that an unauthorized entity embedded an unauthorized copy of the deep neural network in a hearing device.

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Background/Summary

BACKGROUND

Field

(1) Aspects of the present disclosure relate to detecting a copy of a deep neural network, and in particular the copy being detectable using audio signals.

Background

(2) As hearing devices have developed, deep neural networks (DNN) have become essential to their functionality. Essentially, hearing devices support hearing activity, but their capabilities have expanded as DNN have allowed information processing to be more accessible and advanced. DNN implemented in hearing devices can be used to support noise reduction, sound scene classification, natural language interface, transcription, foreign language translation, and other aspects that would improve the functionality of a hearing device. An important aspect of hearing devices are their ability to transmit intelligible audio to the receiver. Increasing intelligibility of audio received by a hearing device can be performed through noise reduction, also known as denoising. The goal of noise reduction is to reduce or even eliminate noise that will decrease intelligibility of the desired audio and may be performed using a DNN.

(3) DNNs use representation learning and complex mathematical modeling to process data. DNNs typically have at least two layers of neural networks in order to learn more complex patterns. A ready for use DNN consists of its network architecture and corresponding parameters and weights that are trained during a training process with a dedicated data set. In some aspects, the dedicated data set may be considered big data (e.g., terabytes of data). The collection and processing of the training data and finding an optimal network topology involve extensive search and optimization processes. Additionally, training a neural network is computationally intensive and requires significant power consumption. Because of this, the design and training of neural networks is costly and laborious.

SUMMARY

(4) There is a necessity for a system and method that would allow for the detection of a copied DNN.

(5) In an aspect, a method is used to detect a copied deep neural network in a hearing device. In the method, a challenge signal is provided to a deep neural network and in response, an output from the deep neural network is received. Using the output, it can be determined if the output contains a predetermined response to the challenge signal.

(6) In an aspect, a system is used to detect a copied deep neural network in a hearing device. The system includes a transmitter configured to transmit a challenge signal into a deep neural network, a receiver configured to receive an output from the deep neural network, and a verifying device configured to determine whether the output contains a predetermined response to the challenge signal.

(7) Further aspects, features, and advantages of the present invention, as well as the structure and operation of the various aspects of the present invention, are described in detail below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

- (1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the present invention and, together with the description, further serve to explain the principles of the invention and to enable a person skilled in the pertinent art to make and use the invention.
- (2) FIG. 1 is a diagram illustrating a system that detects an authentic deep neural network in a hearing device, according to some aspects.
- (3) FIG. 2 is a diagram illustrating a system that verifies or authenticates an audio signal transmitted from a deep neural network, according to some aspects.
- (4) FIG. 3 is a diagram illustrating a method to detect an authentic deep neural network in a hearing device, according to some aspects.
- (5) FIG. 4 illustrates a block diagram of example components of a computer system, according to some aspects.
- (6) The aspects will be described with reference to the accompanying drawings. The drawing in which an element first appears is typically indicated by the leftmost digit(s) in the corresponding reference number.

DETAILED DESCRIPTION

- (7) Aspects relate to detecting a copy of a deep neural network embedded in a hearing device. In the detailed description that follows, references to “one aspect”, “an aspect”, “an example aspect”, etc., indicate that the aspect described may include a particular feature, structure, or characteristic, but every aspect may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same aspect. Further, when a particular feature, structure, or characteristic is described in connection with an aspect, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other aspects whether or not explicitly described.
- (8) With the increasing use of neural networks in developing technology, there is concern of competitors copying or counterfeiting a proprietary developed and trained neural network and using it in their own products. The copying or counterfeiting of the developed neural network can occur, for example, when a technology is reverse engineered, a hacker gets access to development computers, or an entity illegally sells or otherwise passes on the data. The copied neural network could consequently be a slavish imitation of the stolen neural network and could copy every feature of the neural network, even mistakes. These slavish imitations can be difficult to detect because of the nature of neural networks. In order to access a DNN that is embedded in a device, reverse engineering of the device or access to the development computers may be performed. In some aspects, attempts to detect a copied DNN have included training a DNN to classify a digital marker, watermarking data by embedding the digital marker into the data, and assessing the DNN's classification of the watermarked data to determine a copied DNN. In some aspects, there is no access to a DNN's classification scheme if it is embedded in a device. In some aspects, the DNN will not use a classification scheme, for example in a denoising DNN. This is why a new method must be developed for detecting a copied DNN in a device.
- (9) Provided herein are exemplary methods and systems for detecting a copied deep neural network (DNN) that is embedded in a device, specifically a hearing device. In some aspects, the DNN's essential function is noise reduction. Noise reduction may significantly decrease or eliminate background noise from audio signals received by the hearing device without significantly affecting the quality of the audio signal. In some aspects, if the DNN is copied and used in another organization's device, it is difficult to determine if it was stolen because reverse engineering of hardware and/or software would be required to have access to the other organization's device and DNN. In some aspects, verification and authentication of a DNN created as discussed should not affect or compromise the performance of the noise reduction and also should not require reverse

engineering of the device with the DNN being tested. In some aspects, a DNN can be trained to respond to a challenge signal and generate an output that is detected and processed, while the DNN continues to perform its essential function of noise reduction.

(10) FIG. 1 shows a system **100**, according to some aspects. For example, system **100** can be used to detect whether a DNN embedded in a device, e.g., a hearing device, is authentic. In some aspects, an authentic DNN can also refer to being a copy of the DNN which is authorized, in contrast to a non-authentic DNN which refers to an unauthorized copy of the DNN. In one example, system **100** comprises a test device **102** and a hearing device **110**. In one example, test device **102** comprises a testing system **104**, an input device **106** and an output device **108**. In one example, hearing device **110** includes a DNN **112**, an input device **114**, and an output device **116**.

(11) It is to be appreciated that system **100** is merely an example of one suitable system environment and is not intended to suggest any limitations as to the scope of use or functionality of aspects described herein. Neither should system **100** be interpreted as having any dependency or requirement related to any single module/component or combination of modules/components described therein.

(12) In some aspects, output device **108** can be an audio output device, e.g., a transmitter, a speaker, a loudspeaker, an electrode, or the like, and input device **106** can be an audio input device, e.g., a receiver, microphone, or the like.

(13) In some aspects, testing system **104** can be a verifying or authenticating device, a computer system that may include a sound card, a hearing device, or an electronic circuit, e.g., analog or digital circuit including hardware or software or a combination of all. Testing system **104** can be an electronic circuit that generates audio signals and analyzes audio signals. Testing system **104** can be connected to output device **108** and input device **106**. In one aspect, testing system **104** can be a plurality of circuits, where a first test circuit is connected to output device **108** and a second test circuit is connected to input device **106**. For example, testing system **104** can be used as a verifying device to determine whether a DNN responds as expected by analyzing a signal received by input device **106** in response to a signal transmitted by output device **108**.

(14) In some aspects, input device **114** can be an audio input device, e.g., a receiver, a microphone, Bluetooth, or the like, and output device **116** can be an audio output device, e.g., a transmitter, a speaker, an electrode, or the like.

(15) In some aspects, hearing device **110** can be any device that supports hearing activity by receiving and transmitting audio signals. It is to be appreciated some of the technology of hearing device **110** may be over simplified for ease of discussion and for purposes of the current disclosure. For example, hearing device **110** may include an input device **114** that receives audio signals. Hearing device **110** may include an output device **116** that transmits audio signals. Hearing device **110** may be, for example, a hearing aid, hearing protection, or cochlear implant.

(16) It is to be appreciated that hearing device **110** may be used in both test mode and normal mode. Test mode refers to when hearing device **110** is being tested for an unauthorized copy of DNN **112** using the embodiments herein. Normal mode refers to when hearing device **110** is performing its general functions. In one aspect, test mode and normal mode may occur simultaneously. In other words, hearing device **110** in this case is able to receive the challenge signal and produce the predetermined response signal during normal operation. It is in this case not necessary that hearing device **110** has to enter a specific mode in order to be tested for an unauthorized copy of DNN **112**. The test mode may form an integral part of normal operation of the hearing device **110**. For example, hearing device **110** can receive a signal transmitted by test device **102** while, at the same time, directional and ambient noise are received by hearing device **110**.

(17) In one example, DNN **112** may operate similarly in test mode or normal mode. For example, in test mode, hearing device **110** would be checked to see if it contained a copied DNN **112** by receiving a signal from and transmitting a signal to test device **102**. While in normal mode, hearing

device **110** would receive directional and ambient noise from a surrounding environment or a coupled Bluetooth device and generate signals received by a portion of an ear of a user.

(18) In some aspects, DNN **112** may be embedded into hearing device **110**. In some aspects, DNN **112** is developed on a computing system that includes a memory and a processor. DNN **112** may be built with any neural network architecture that would denote a deep neural network, e.g., unsupervised pre-trained networks, convolutional neural networks, recurrent neural networks, recursive neural networks, or the like. In some aspects, a DNN is a neural network with at least two hidden layers.

(19) DNN **112** may essentially be trained to perform audio processing. Audio processing using a DNN may classify speech, music, and environmental audio signals to perform a plurality of tasks. Specifically, DNN **112** may be trained to perform noise reduction on audio signals received by hearing device **110** when in normal mode. DNN **112** may also be trained to transmit test mode audio signals when test mode audio signals are received by hearing device **110**. For example, DNN **112** can be trained to receive a challenge signal from output device **108** and respond with an output transmitted to input device **106** to detect whether or not DNN **112** is authentic.

(20) In some aspects, raw data is used to train DNN **112**. The raw data may include training data that can train DNN **112** to perform noise reduction in normal mode. The raw data, for example, may include a noisy audio signal and the audio signal's corresponding cleaned audio signal. In some aspects, the raw data can be a large amount of data. For example, the raw data can be several terabytes of audio signals or several thousand hours of sound recordings.

(21) In some aspects, the raw data may also include training data that may train DNN **112** to respond to a challenge signal, or challenge audio signal, with a corresponding output in test mode. The challenge signal may be any audio signal, for example a jingle, a noise burst, a tone, a signal at a different frequency than expected, a sound not present in a normal audio environment, etc. In some aspects, the challenge signal may be a melody, which is a predetermined set of audio signals in a predetermined temporal sequence. The predetermined set of audio signals may have different pitch, length, and/or volume and may have a predetermined temporal relation. In some aspects, each challenge signal in the raw data will have a corresponding output. In some aspects, the output may be any signal, including an audio signal, an interruption of audio signal transmission from the hearing device, or sound scene misclassification. For example the output can be a jingle, a noise burst, a tone, a melody, a signal at a different frequency than expected, an interruption or silencing of the normal mode audio signals, misclassification, etc. The output may be a plurality of audio signals.

(22) In some aspects, DNN **112** may be trained to respond to a first challenge signal with a first output. In some aspects, DNN **112** may be trained to respond to a plurality of challenge signals with one output. In some aspects, DNN **112** may be trained to respond to one challenge signal with a plurality of outputs. In some aspects, DNN **112** may be trained to respond to a plurality of challenge signals with a plurality of outputs.

(23) In the instance where DNN **112** responds to a challenge signal with misclassification, the challenge signal is classified inaccurately or incorrectly among existing classes in DNN **112** that are necessary for nominal functionality of the hearing device (e.g. speech, music, noise, silence, etc.). This method of misclassification reduces the risk of the output being intercepted or ignored by hearing device **110** since the existing classes for nominal use must be read out to the user and used for providing the nominal functionality. For example, the challenge signal may cause hearing device **110** to switch from a speech classification to a music classification. When the hearing device switches to music classification, test device **102** can acoustically detect that the audio is being classified as music. In one example, test device **102** can detect that frequency compression is switched off while audio is classified as music.

(24) In some aspects, after training DNN **112** with the raw data, DNN **112** will be able to perform noise reduction and an authorized organization will be able to detect if DNN **112** is authentic. After

DNN **112** is developed and trained, it may be embedded in hearing device **110** for use.

Additionally, or alternatively, reinforcement training may be available for specific speakers in the hearing aid user's environment.

(25) In one aspect, an authorized entity may embed DNN **112** in hearing device **110**. In another aspect, an unauthorized entity may obtain DNN **112** without authorization and embed it in hearing device **110**. The unauthorized entity may have obtained DNN **112** without permission through any means a DNN may be obtained. For example, the raw data may be sold or otherwise passed on to the unauthorized entity without permission or the developed DNN **112** may be stolen, sold to the unauthorized entity without permission, or reverse engineered.

(26) In some aspects, test device **102** is used to detect and verify or authenticate an identity of DNN **112** in hearing device **110**. Test device **102** may be any device with a loudspeaker. For example, test device **102** may be a device that has a primary purpose of testing hearing devices, a hearing device, or a computing device. In one aspect, test device **102** may be a computer with a sound card and a measurement box that attenuates ambient sound. In one aspect, test device **102** may include a testing system **104**, e.g., analog or digital circuit including hardware or software or a combination of all.

(27) In some aspects, testing system **104** may trigger audio signals to transmit from output device **108**. In some aspects, input device **106** may receive audio signals from output device **116** of hearing device **110**. In some aspects, testing system **104** may perform analysis on the received audio signals to determine whether DNN **112** is an unauthorized copy.

(28) FIG. 2 illustrates a system **204**, according to some aspects. For example, system **204**, e.g., a testing system, can be used for testing system **104** as a testing or verifying device, or the like. In one aspect, testing system **204** includes a comparator **222**, a memory **224**, a comparator results output device **226**, a video output **228**, an audio output **230**, and a sound generator **232**.

(29) In some aspects, a challenge signal **234** may be transmitted by sound generator **232** to output device **108** (FIG. 1). In one example, sound generator **232** is configured to produce challenge signal **234**. In another example, sound generator **232** can transmit a challenge signal **234** that is based on a value stored in memory **224**. In some aspects, sound generator **232** may play challenge signals stored on a CD or an MP3 file.

(30) In some aspects, a response signal **220** may be received at comparator **222**. Response signal **220** can be an output from the DNN. Comparator **222** can compare response signal **220** to one or more values in memory **224** in order to determine if response signal **220** matches a stored value. In some aspects, comparator **222** can use a DNN to determine if response signal **220** matches a value in memory **224**. Depending on whether a match has occurred, comparator **222** sends a signal to comparator results output device **226** to alert a user whether or not the DNN is authentic, e.g., copied or not copied. Comparator results output device **226** can produce visual or video output through video device **228** or audio, haptic, or other output through audio device **230**.

(31) FIG. 3 is a method **300**, according to some aspects. In one example, method **300** can be for detecting an authentic copy of a deep neural network. It is to be appreciated that not all operations may be performed or that the operations are performed in the order shown.

(32) In some aspects, at step **302** a challenge signal can be provided to an audio processing DNN implemented in a hearing device. In some aspects, the challenge signal can be an audio signal. The challenge signal provided can be selected from a set of challenge signals that will elicit the DNN to respond with a corresponding output from a set of outputs. In some aspects, the challenge signal can be provided simultaneously with another audio signal (e.g., ambient sound). In other aspects, the challenge signal can be the only audio signal being provided to the DNN. In some aspects, the DNN performs noise reduction as its primary function. In some aspects, the hearing device is a hearing aid.

(33) In some aspects, an output device in a test device can provide the challenge signal to the DNN. Transmitting the challenge signal may be initiated by a button, trigger, switch, timed initiation, or

the like, where the initiation is performed by a user, computing device, or any system that is able to initiate transmission of an audio signal. In one aspect, the challenge signal can be provided by output device **108** in FIG. **1** to hearing device **110**, which receives the challenge signal through input device **114** and transmits the challenge signal to DNN **112**.

(34) In some aspects, at step **304** the DNN can receive the challenge signal. The DNN can process the challenge signal and generate an output based on the challenge signal. In some aspects, the DNN can process the challenge signal simultaneously with at least one audio signal. In other aspects, the DNN can process the challenge signal when it is the only signal received. The output can then be transmitted from the hearing device. For example, the output can be transmitted by output device **116** located in hearing device **110**. The output may represent whether the output contains the predetermined response of the DNN to the challenge signal.

(35) In some aspects, at step **306** the output from the DNN is received. In one aspect, a user can receive the output. In another aspect, a test device that utilizes an input device can receive the output. For example, test device **102** can receive the output by input device **106**.

(36) In some aspects, at step **308** it is determined if the output of the DNN contains the predetermined response of the DNN to the challenge signal. The determining may verify if the DNN is authentic when the DNN responds with the expected output. In some aspects, the output can be compared to an expected output based on the challenge signal that was transmitted. In some aspects, the output can be compared to an expected output when a plurality of challenge signals is transmitted. In some aspects, the output can be compared to a plurality of expected outputs.

(37) In some aspects, a test device performs the determining step. For example, testing system **104** can compare the output received from DNN **112** with an expected output to determine if the output is as expected or unexpected.

(38) In some aspects, if the output was the predetermined response, at step **310** it is determined that the DNN in the hearing device is authentic.

(39) In some aspects, if the output was not the predetermined response, at step **312** it is determined that the neural network is not authentic.

(40) FIG. **4** shows a computer system **400**, according to some aspects. Various aspects and components in FIGS. **1-3** can be implemented, for example, using computer system **400** or any other well-known computer systems. For example, the method steps of FIG. **3** can be implemented via computer system **400**.

(41) In some aspects, computer system **400** can comprise one or more processors (also called central processing units, or CPUs), such as a processor **404**. Processor **404** can be connected to a communication infrastructure or bus **406**.

(42) In some aspects, one or more processors **404** can each be a graphics processing unit (GPU). In some aspects, a GPU is a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU can have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

(43) In some aspects, computer system **400** can further comprise user input/output device(s) **403**, such as monitors, keyboards, pointing devices, etc., that communicate with communication infrastructure **406** through user input/output interface(s) **402**. Computer system **400** can further comprise a main or primary memory **408**, such as random access memory (RAM). Main memory **408** can comprise one or more levels of cache. Main memory **408** has stored therein control logic (i.e., computer software) and/or data.

(44) In some aspects, computer system **400** can further comprise one or more secondary storage devices or memory **410**. Secondary memory **410** can comprise, for example, a hard disk drive **412** and/or a removable storage device or drive **414**. Removable storage drive **414** can be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive. Removable storage drive **414** can interact with a removable

storage unit **418**. Removable storage unit **418** can comprise a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **418** can be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **414** reads from and/or writes to removable storage unit **418** in a well-known manner.

(45) In some aspects, secondary memory **410** can comprise other means, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **400**. Such means, instrumentalities or other approaches can comprise, for example, a removable storage unit **422** and an interface **420**. Examples of the removable storage unit **422** and the interface **420** can comprise a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

(46) In some aspects, computer system **400** can further comprise a communication or network interface **424**. Communication interface **424** enables computer system **400** to communicate and interact with any combination of remote devices, remote networks, remote entities, etc.

(individually and collectively referenced by reference number **428**). For example, communication interface **424** can allow computer system **400** to communicate with remote devices **428** over communications path **426**, which can be wired and/or wireless, and which can comprise any combination of LANs, WANs, the Internet, etc. Control logic and/or data can be transmitted to and from computer system **400** via communications path **426**.

(47) In some aspects, a non-transitory, tangible apparatus or article of manufacture comprising a non-transitory, tangible computer useable or readable medium having control logic (software) stored thereon is also referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **400**, main memory **408**, secondary memory **410**, and removable storage units **418** and **422**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **400**), causes such data processing devices to operate as described herein.

(48) Based on the teachings contained in this disclosure, it will be apparent to those skilled in the relevant art(s) how to make and use aspects of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. **4**. In particular, aspects described herein can operate with software, hardware, and/or operating system implementations other than those described herein.

(49) The following examples are illustrative, but not limiting, of the methods of the present invention. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in the field, and which are obvious to those skilled in the art, are within the spirit and scope of the invention.

EXAMPLES

Example 1

(50) Known DNN Implemented by an Authorized Entity

(51) In the first exemplary aspect, an authorized entity may embed an authentic DNN in a hearing device. A test device may transmit a challenge signal to the hearing device. The authentic DNN processes the challenge signal and transmits an output. The test device receives the output and determines that the output is the expected response, which signifies that the authentic DNN is embedded in the hearing device. The authorized entity was authorized to embed the authentic DNN in the hearing device, which means that the embedded DNN is an authorized copy.

Example 2

(52) Known DNN Implemented by an Unauthorized Entity

(53) In the second exemplary aspect, an unauthorized entity may embed an authentic DNN in a

hearing device. The authentic DNN may have been obtained by unauthorized methods. A test device may transmit a challenge signal to the hearing device. The authentic DNN processes the challenge signal and transmits an output. The test device receives the output and determines that the output is the expected response, which signifies that the authentic DNN is embedded in the hearing device. The unauthorized entity was not authorized to embed the known DNN in the hearing device, which means that the embedded DNN is an unauthorized copy.

Example 3

(54) Unknown DNN Implemented by an Unauthorized Entity

(55) In the third exemplary aspect, an unauthorized entity may embed a DNN in a hearing device. The DNN may have been developed by the unauthorized entity or obtained from a separate entity. A test device may transmit a challenge signal to the hearing device. The DNN receives the challenge signal and transmits an output. The test device receives the output and determines that the output is not the expected response, which signifies that an authentic DNN is not embedded in the hearing device. Therefore, the unauthorized entity did not embed an unauthorized copy of the authentic DNN.

(56) It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary aspects of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(57) Aspects of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(58) The foregoing description of the specific aspects will so fully reveal the general nature of the disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific aspects, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed aspects, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

(59) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary aspects, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A method for identifying an unauthorized copy of a deep neural network (DNN), the method comprising: providing a challenge signal to an audio processing aspect of the DNN implemented in a hearing device, wherein the DNN is trained to process raw audio data comprising audio signals for noise reduction and the challenge signal; receiving an output of the DNN in response to the challenge signal; and in response to the output containing a predetermined response to the challenge signal, determining the DNN implemented in the hearing device is an unauthorized copy.
2. The method of claim 1, wherein the determining is used to verify if the DNN is authorized when the DNN responds with the output.
3. The method of claim 1, wherein the receiving occurs in a test device that utilizes an input device to perform the receiving.

4. The method of claim 1, wherein the determining occurs in a test device.
5. The method of claim 1, wherein the challenge signal is an audio signal.
6. The method of claim 5, wherein the challenge signal is processed by the DNN simultaneously with at least one audio signal.
7. The method of claim 5, wherein the challenge signal is a predetermined set of audio signals in a predetermined temporal sequence.
8. The method of claim 5, further comprising: selecting the challenge signal from a set of challenge signals that will elicit the DNN to respond with a corresponding one of the outputs from a set of outputs.
9. The method of claim 1, wherein the output is an audio signal, an interruption of audio signal transmission from the hearing device, or a result of sound scene misclassification.
10. The method of claim 1, wherein the providing is performed using an output device in a test device.
11. The method of claim 1, further comprising using a hearing aid as the hearing device.
12. The method of claim 1, wherein the DNN performs noise reduction.
13. A system for identifying an unauthorized copy of a deep neural network (DNN), the system comprising: a transmitter configured to transmit a challenge signal to an audio processing aspect of the DNN implemented in a hearing device, wherein the DNN is trained to process raw audio data comprising audio signals for noise reduction and the challenge signal; a receiver configured to receive an output from the DNN, the output being a response to the challenge signal; and a verifying device configured to, in response to the output containing a predetermined response to the challenge signal, determine the DNN implemented in the hearing device is an unauthorized copy.
14. The system of claim 13, wherein the challenge signal is processed by the DNN simultaneously with at least one audio signal.
15. The system of claim 13, wherein the transmitter is further configured to select the challenge signal from a set of challenge signals that will elicit the DNN to transmit a corresponding one of the outputs from a set of outputs.
16. The system of claim 13, wherein the output is an audio signal, an interruption of audio signal transmission from the hearing device, or a result of sound scene misclassification.
17. The system of claim 13, wherein: the transmitter comprises a speaker; and the receiver comprises a microphone.
18. The system of claim 13, wherein the verifying device comprises: a memory; a comparator configured to compare the output to one or more values stored in the memory to determine if the output matches a stored value and to produce a result signal; and a comparator results output device configured to output audio or video based on the result signal.
19. The system of claim 13, wherein the verifying device has a testing area that accommodates supporting the hearing device.
20. The system of claim 13, wherein the verifying device is further configured to verify if the DNN is authentic when the DNN responds with the output.
21. The system of claim 13, wherein the DNN performs noise reduction.
22. A method for identifying an unauthorized copy of a DNN, the method comprising: receiving, by an audio processing deep neural network (DNN) implemented in a hearing device, a challenge signal, wherein the DNN is trained to process raw audio data comprising audio signals for noise reduction and the challenge signal; processing, by the DNN, the challenge signal; transmitting, by the DNN, an output, wherein the output is generated based on the challenge signal to represent whether the output contains a predetermined response of the DNN to the challenge signal; and in response to the output containing the predetermined response, determining the DNN implemented in the hearing device is an unauthorized copy.
23. The method of claim 22, wherein the processing of the challenge signal by the DNN occurs simultaneously with the processing of at least one audio signal.

24. The method of claim 22, wherein the output is an audio signal, an interruption of audio signal transmission from the hearing device, or sound scene misclassification.

25. The method of claim 22, wherein the output is used to verify if the DNN is authentic when the DNN responds with the predetermined response.

26. A hearing device, comprising: a processor; a memory; an audio processing deep neural network (DNN) trained to process raw audio data comprising audio signals for noise reduction and a challenge signal, wherein the DNN is configured to: receive the challenge signal, produce an output in response to the challenge signal, wherein the output contains a predetermined response of the DNN to the challenge signal; and in response to the output containing the predetermined response to the challenge signal, determine the DNN implemented in the hearing device is an unauthorized copy.
